

Lab10: Structural Bioinformatics 1

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Background

The main repository of high-resolution structural data on biomolecules is called “**Protein Data Bank**” (PDB).

PDB Statistics

What is in the PDB in terms of molecules type and structure determination method?

Here we read the data from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					

1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it, so we need to fix this.

Two functions that can help are `sub()` and `as.numeric()`.

```
## We want to get rid of commas or (sub out) commas
sub(",", "", x=pdb$X.ray)
```

```
[1] "180758" "10488" "9205" "3154" "178" "11"
```

Or using the `a.numeric()` to convert the characters into numeric values.

```
as.numeric(sub(",", "", x=pdb$X.ray))
```

```
[1] 180758 10488 9205 3154 178 11
```

```
x <- pdb$X.ray
tmp <- sub(",", "", x=pdb$X.ray)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this

```
rm.comma <- function(x) {
  tmp <- sub(",", "", x=pdb$X.ray)
  sum(as.numeric(tmp))
}
```

```
rm.comma(pdb$Total)
```

```
[1] 203794
```

```
rm.comma(pdb$EM)
```

```
[1] 203794
```

```
rm.comma(pdb$X.ray)
```

```
[1] 203794
```

We could also use a different import functions for this CSV file that speaks American (i.e. deals with comma in numbers in a comma separated file).

```
library(readr)
```

```
stats <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
n.tot <- sum(stats$Total)
```

```
n.em <- sum(stats$EM)
```

```
n.xray <- sum(stats$`X-ray`)
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
(n.xray/n.tot)*100
```

```
[1] 80.48577
```

```
(n.em/n.tot)*100
```

```
[1] 13.38046
```

```
pdb$Total[1]
```

```
[1] "217,375"
```

Q2: What proportion of structures in the PDB are protein?

The 97.9% of the total PDB are proteins.

```
n.protein <- sum(stats$Total[grep("Protein", stats$`Molecular Type`)])  
(n.protein / n.tot) * 100
```

```
[1] 97.91868
```

Key-point We have a very very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

Skipped!

Visualizing PDB Data

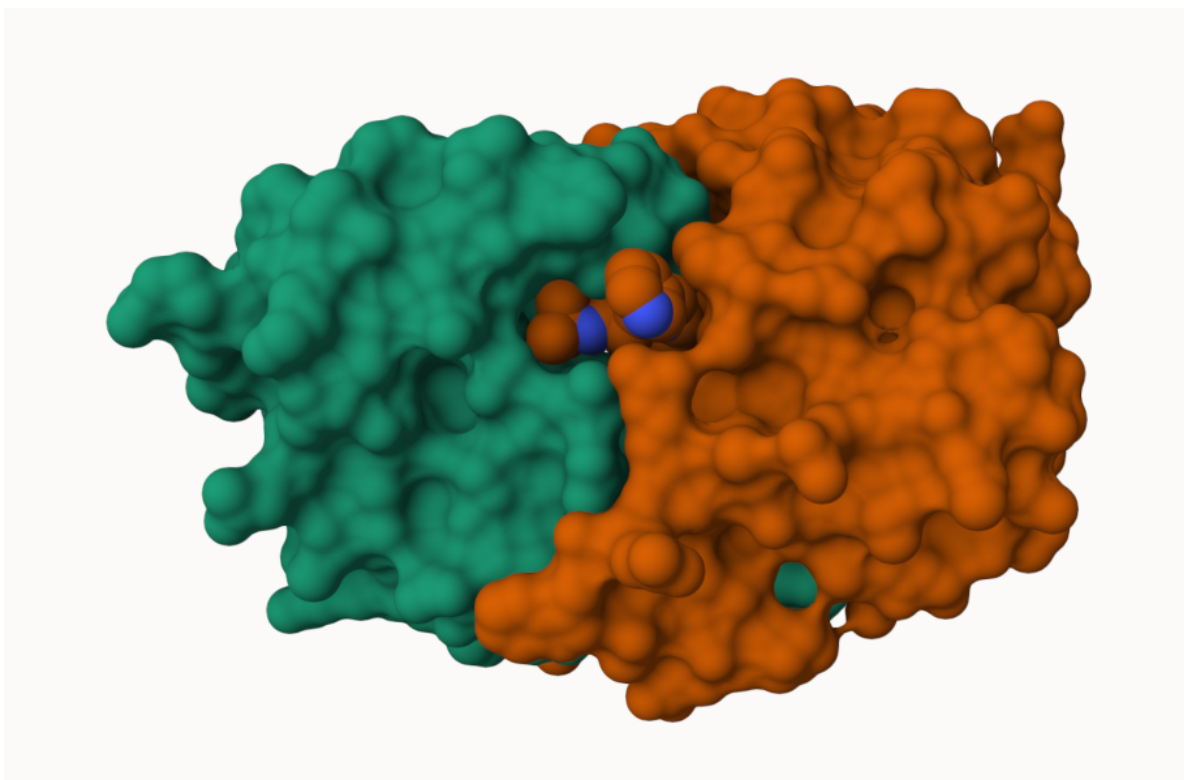
Main stand alone web version with all features is at, <https://molstar.org/viewer/>

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

The structure shows the space fill picture of the water molecule so it is shown as only one molecule.



Figure 1: The HIV-protease enzyme is a homodimer with two chains



> Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The highlighted red ball represents the water molecule.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.



Figure 2: Spacefill display of catalytic ASP25 amino acids and key binding site water molecule

```
library(bio3d)
pdb <- read.pdb("1HSG")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1HSG")
```

```
Total Models#: 1
  Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in the pdb file.

Q8: Name one of the two non-protein residues?

The two non protein residues are HOH water and MK1.

Q9: How many protein chains are in this structure?

There are 2 protein chains in the structure.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom"    "xyz"     "seqres"  "helix"   "sheet"   "calpha"  "remark"  "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>


```
5 <NA>      C  <NA>
6 <NA>      C  <NA>
```

```
library(bio3dview)
library(NGLViewerR)

pdb <- read.pdb("1HSG")
view.pdb(pdb)
```

```
sele <- atom.select(pdb, resno = 25)

view.pdb(pdb, cols = c("navy", "teal"),
         highlight = sele,
         highlight.style = "spacefill") |>
  setRock()
```

Q. Create a custom view highlighting the active site ASP (**resno=25**) the two chains in your choice of color and the ligand all on a custom color background.

```
active.site <- atom.select(pdb, resno = 25)

view.pdb(pdb,
         cols = c("blue", "red"),
         highlight = active.site,
         highlight.style = "spacefill",
         backgroundColor = "pink") |>
  setRock()
```

Predict the flexibility of a given structure

Lets do a normal mode analysis NMA to predict the flexibility of a give pdb file.

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, `rm.alt=TRUE`

A quick structure summary

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM TAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

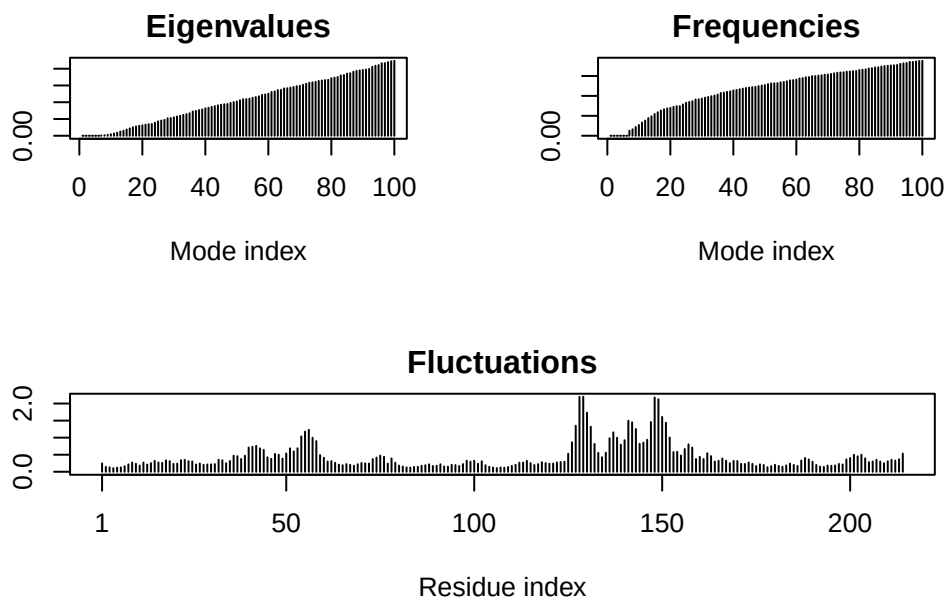
```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

```
m <- nma( adk )
```

```
Building Hessian... Done in 0.012 seconds.
```

```
Diagonalizing Hessian... Done in 0.262 seconds.
```

```
plot(m)
```



View the results with

```
view.nma(m)
```

Q10. Which of the packages above is found only on BioConductor and not CRAN?

Bio3dview is only found in bioConductor not CRAN.

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
mktrj(m, file="adk_m7.pdb")
```

Q11. Which of the above packages is not found on BioConductor or CRAN?:

NGLViewer is not found in bioConductor or CRAN because it is installed from GitHub.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True!

Comparative Analysis of the ADK family

Our first step is to find a sequence for this family. We will use the database ID “lake_A” here

```
id <- "lake_A"
aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```

      1      .      .      .      .      .      60
pdb|1AKE|A MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

     121      .      .      .      .      .      180
pdb|1AKE|A VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

```
+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids long with no gap.

```
blast <- blast.pdb(aa)
```

```
Searching ... please wait (updates every 5 seconds) RID = ZUD13K79016
```

```
...
```

```
Reporting 96 hits
```

```
head(blast$hit.tbl)
```

	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_6577651	1AKE_A	100.000	214	0	0	1
2	Query_6577651	8BQF_A	99.533	214	1	0	1
3	Query_6577651	4X8M_A	99.533	214	1	0	1
4	Query_6577651	6S36_A	99.533	214	1	0	1
5	Query_6577651	9R6U_A	99.533	214	1	0	1
6	Query_6577651	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evaluate	bitscore	positives	mlog.evaluate	pdb.id	acc
1	214	1	214	1.81e-156	432	100.00	358.6099	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.76e-156	432	100.00	357.6430	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

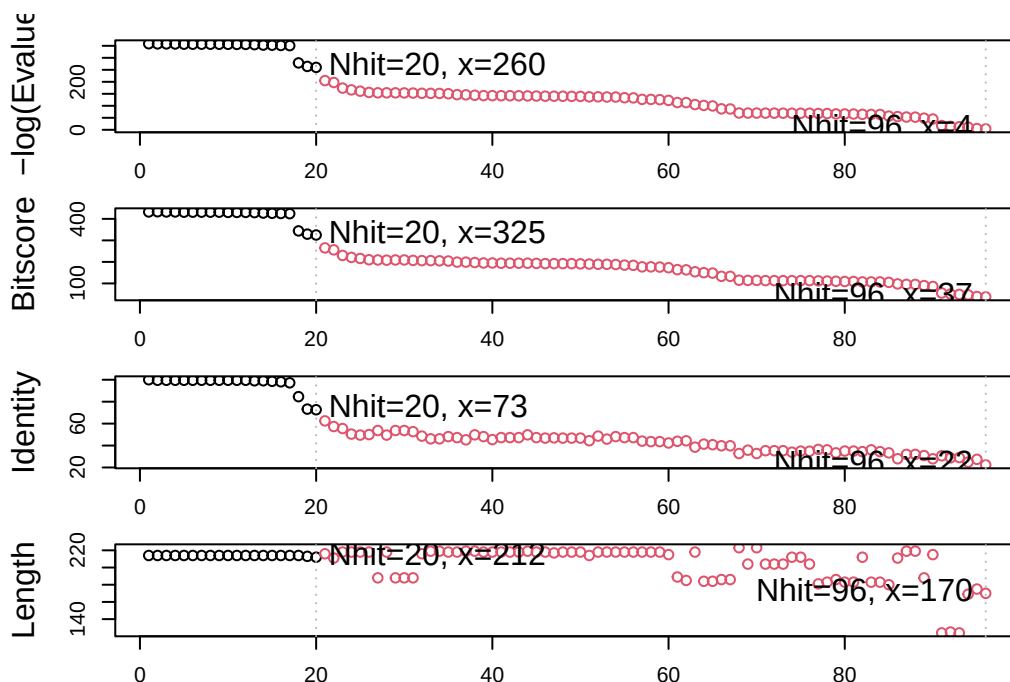
```
hits <- NULL
```

```
hits$ pdb.id <- c('1AKE_A', '6S36_A', '6RZE_A', '3HPR_A', '1E4V_A', '5EJE_A', '1E4Y_A', '3X2S_A', '6H
```

```
plot(blast)
```

```
* Possible cutoff values: 260 3
    Yielding Nhits: 20 96
```

```
* Chosen cutoff value of: 260
    Yielding Nhits: 20
```



```
hits$ pdb.id
```

```
[1] "1AKE_A" "6S36_A" "6RZE_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A"
[9] "6HAP_A" "6HAM_A" "4K46_A" "3GMT_A" "4PZL_A"
```

```
files <- get.pdb(hits$ pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$ pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

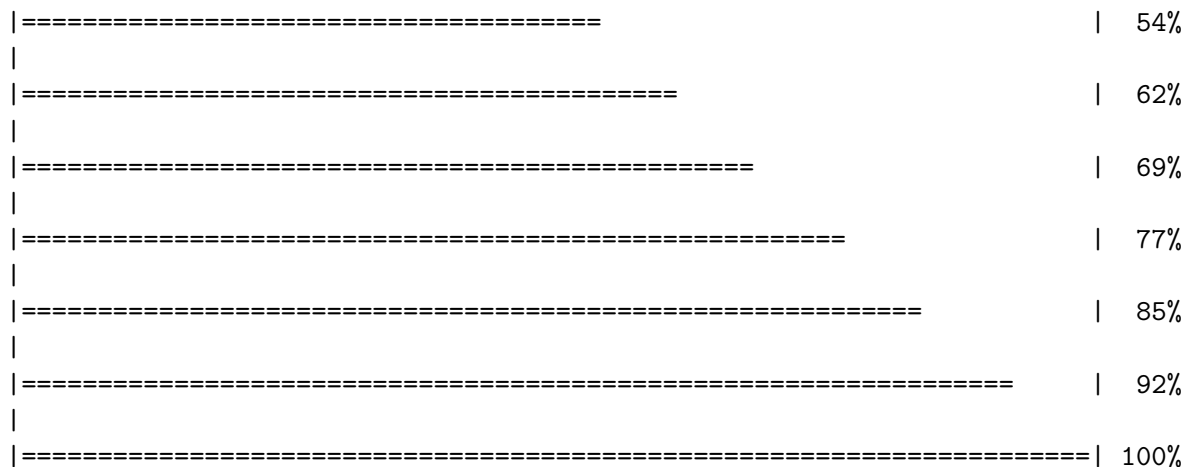
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3GMT.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download

	0%
=====	8%
=====	15%
=====	23%
=====	31%
=====	38%
=====	46%



Align and superpose all these ADK structures.

```
# Align related PDBs
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/3GMT_A.pdb
pdbs/split_chain/4PZL_A.pdb
  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...
```

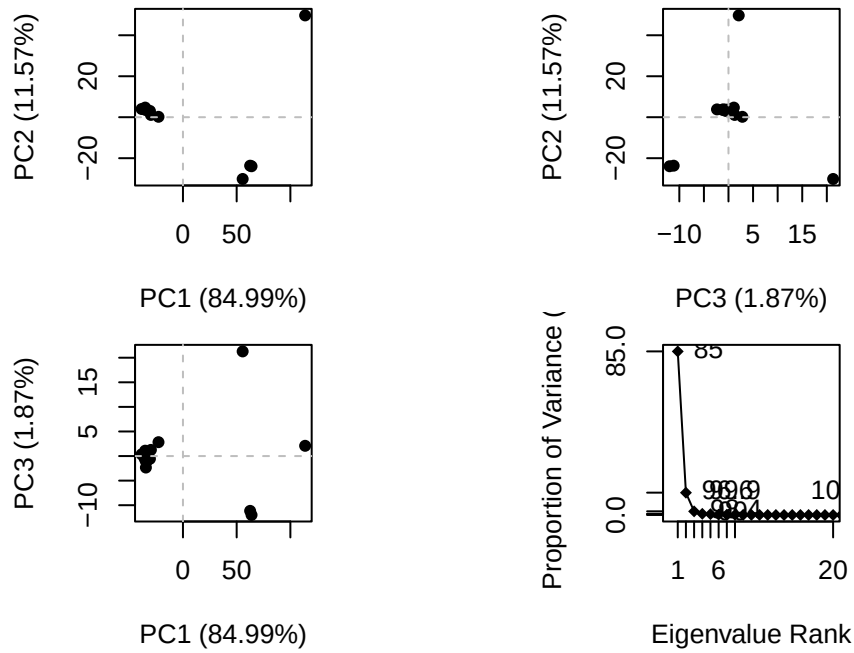

Extracting sequences

```
pdb/seq: 1  name: pdbs/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2  name: pdbs/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3  name: pdbs/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4  name: pdbs/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5  name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 6  name: pdbs/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7  name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 8  name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 9  name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 10 name: pdbs/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11 name: pdbs/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12 name: pdbs/split_chain/3GMT_A.pdb
pdb/seq: 13 name: pdbs/split_chain/4PZL_A.pdb
```

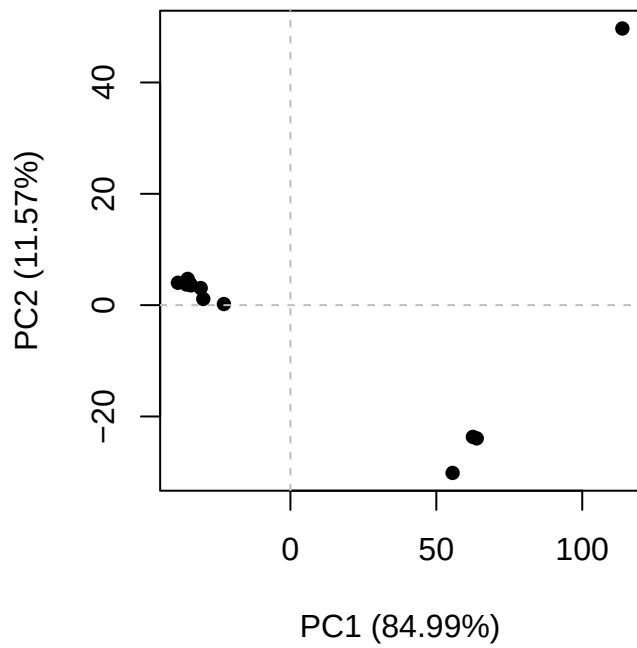
```
library(bio3dview)
view.pdbs(pdbs)
```

PCA of all this structural data (x, y, and z atom coordinates):

```
pc <- pca(pdbs)
plot(pc)
```



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```
view.pca(pc)
```

```
mktrj(pc, file="pca.pdb")
```